

QCalc Geometry Bridge – Python Solver UQFF Integration

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2026-04-13

PAPER_1067: QCalc Geometry Bridge — Python Solver UQFF Integration

Abstract

We document the QCalc geometry bridge connecting QCalc.py numerical solvers to the UQFF framework. The bridge maps QCalc geometric quantities (Christoffel symbols, Riemann tensor, geodesic deviation) to UQFF buoyancy fields via $g_{Ug_sum} = \text{sum}(Ug1 + Ug2 + Ug3 + Ug4)$ with β_i weighting. Validation: QCalc $g_{Ug_sum}(\text{Sun}) = 276.8 \text{ m/s}^2$ vs UQFF $g_{base} = 274.0 \text{ m/s}^2$ (1.0% agreement).

1. Key Equations

- $g_{Ug_sum} = \sum_{i=1}^4 U_{g,i} \cdot \beta_i = 276.8 \text{ m/s}^2 \text{ (Sun)}$
- QCalc-UQFF agreement: 1.0% for solar surface gravity

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class QCalcGeometryBridgeCalculator.

References

- PAPER_1066

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: computational (solver bridge)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = 0$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> computational (solver bridge) -> phonon coupling -> UQFF prediction -> $F_{U,Bi,i}$ unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed